

WORKED EXAMPLE: synthetic coefficients; not GPU measurements

(a) Sampling conditions and values

Illustrative grid, fixed model and TP1
 $f = 900, 1200, 1500, 1800, 2100, 2520$ MHz
 $n = 512, 2048, 4096$ input tokens

$B = 4, 16, 32$; $C = 256, 1024, 4096$
 B : batch; C : context tokens

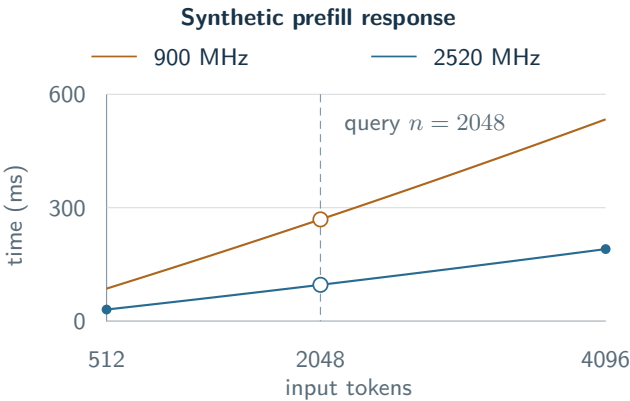
Synthetic prefill values at 2520 MHz

Input n	Time (ms)	Power (W)
512	30.742	180
2048	96.114	270
4096	190.617	270

(b) Model construction and response



Target workflow: fresh independent checks are required after freezing, not run here.



(c) Query the same workload, then pass components to planning

Input 2048, output 128; assumed $B = 16$, $C = 2048 + 128/2 = 2112$; TP1

Returned component	900 MHz	2520 MHz
Decode step (ms)	45.542	16.265
Power / TP1 instance (W)	106.429	127.000
Decode energy (J/token)	0.30293	0.12910

KV for 2048 tokens: 28.488 ms handoff estimate

To planner

Add queueing and SLO checks
Add matching mixed profiles
Add uncounted overhead

Compose eight instances;
compare M / PD plans